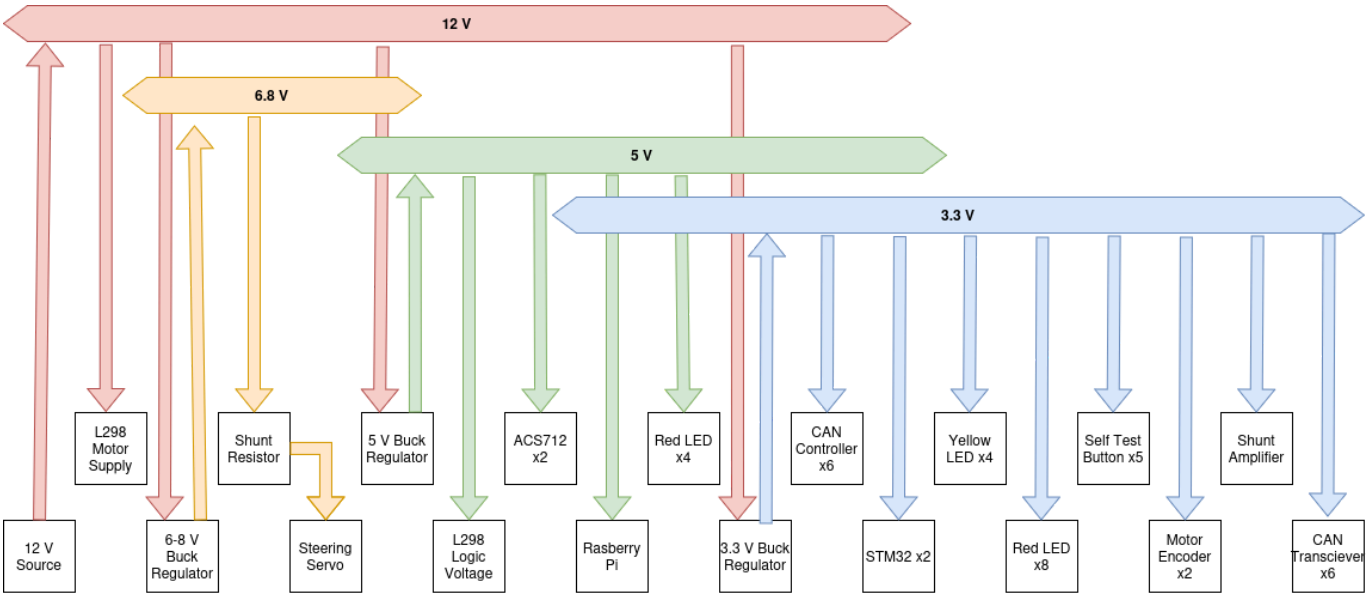
1.1 Power Source And Ratings

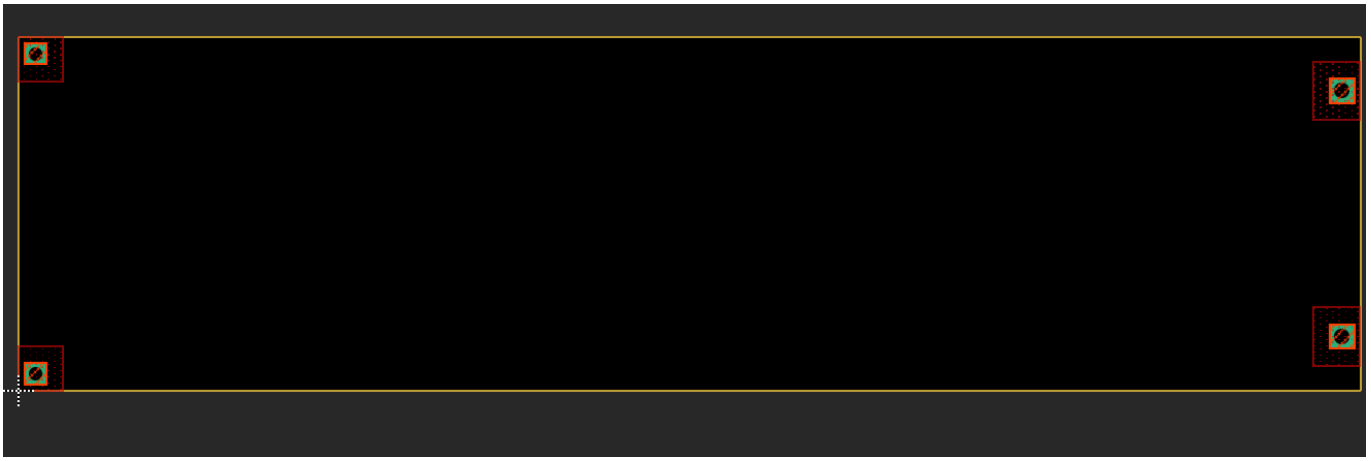


Power Rail	Contributing Factors	Peak Current	Avg Current	Worst-Case Voltage
3.3 Volts	STM32 (36.3 mA max, 31.1 mA typical) x 2, CAN Controller (10 mA max, 10 mA typ) x 6, CAN Transciever (17 mA max, 10 mA typ) x 6, Yellow LED (10.8 mA max, 0 typ) x 4, Red LED (9.2 mA Max, 0 typ) x 8, self test button (0.33 mA max, 0 typ) x 5, INA180 (260 uA max, 197 uA typ), Motor Encoded (estimated 25 mA max, typ) x2	403.31 mA	232.34 mA	3.3 V
5 Volts	L298 Logic (36 mA max, 24 typ), ACS712 (13 mA max, 10 mA typ) x2, Rasberry Pi (3 A max, 1 A typ), RED LED (22.3 mA max, 0 typ) x4	3.151 A	1.044 A	5 V
6.8 Volts	Steering Servo (3 A max, 100 mA typ)	3 A	0.1 A	6.8 V
12 Volts	Motor (3.2 A max, 0.36 typ) x 2, 3V3 Buck Converter (120.6 mA max, 71.0 mA typ), 5V Buck Converter (1.382 A max, 0.472 A typ), 6V8 V Buck Converter (1.89 A max, 64.3 mA typ)	9.79 A	1.327 A	7.105 V

Selected battery source: Tenergy **11616**, a 10S1P Sub-C NiMH battery pack: 12 V nominal, 4.2 Ah (50.4 Wh), 30 A maximum continuous discharge, 113 x 91 x 23 mm, 1.55 lb, with a Standard Tamiya connector. The 30 A rating provides margin over the calculated 9.79 A peak system load. It is an off-board part, but is included in the project BOM. The board-side power input connector must mate with its Standard Tamiya connector (or use a documented Tamiya-to-board harness); do not select an XT60 footprint unless the battery harness is changed accordingly.

For the voltage sag calculation, NiMH cells are conservatively modeled as 50 mOhms per cell. For ten series cells, the worst case voltage is $12 - 9.79 * 10 * .050 = 7.105V$. Measure the selected pack's voltage at load before treating this as a final value. According to the datasheet, the buck regulator has a maximum duty cycle of 97%, meaning that the minimum input voltage needed for the lines to regulate are $V_{in} = 6.8/0.97 = 7.01V$ for the 6.8 volt line, $V_{in} = 5/0.97 = 5.15V$ for the 5 volt line, and $V_{in} = 3.3/0.97 = 3.40V$ for the 3.3 volt line. Thus, at the sagged voltage, all lines should still regulate.

1.2 Mechanical Design



Notes Board has two pairs of 3.2 mm holes in the back, makes more sense to use the inner 2, as well as 2 2.7 mm holes towards the front, gap between back pair is about 46.2 mm from the far end (43 mm from center to center), gap between front pair is about 58.7 mm end to end (56 mm center to center). Distance between the two pairs is 228.8 mm center to center (231.75 mm end to end). JLCPCB design rules state 0.33 mm between copper and board edge, so reasonable board dimensions are 62 mm x 235 mm. Fixing the front left at (0, 0): Front holes (2.7 mm): (0, 0) and (56, 0) Back holes (3.2 mm): (6.5, 228.8) and (49.5, 228.8)

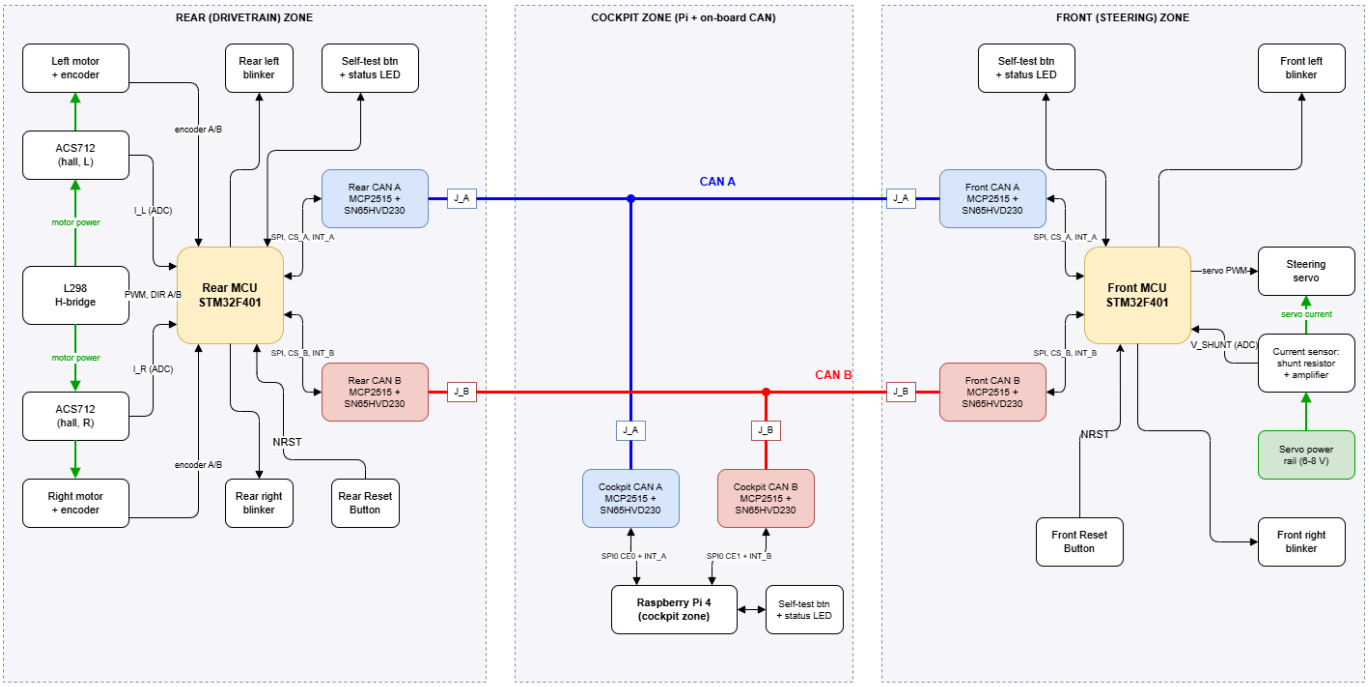
1.3 Power Metering

Sensor	Full-scale current	mA per ADC count	Resistor power rating	Filter corner	Test point names
Motor A: ACS712ELCTR-05B	3.2 (motor max input)	about 1.56 mA per bit	N/A, no external shunt (internal conductor about 1.2 mΩ, roughly 12 mW at 3.2 A)	N/A	[TP12: PWM_L]
Motor B: ACS712ELCTR-05B	3.2 (motor max current)	about 1.56 mA per bit	N/A no external shunt	N/A	[TP13: PWM_R]
Steering servo: shunt + INA180A1 (gain 20)	3 A	about 0.732 per ADC bit (3 A /	3 W	16.931 Hz	[TP14: SERVO_PWM]

Sensor	Full-scale current	mA per ADC count	Resistor power rating	Filter corner	Test point names
		4096 resolution)			

2.0 Block Diagram

Reference hardware block diagram: three zones, dual-redundant CAN



Blue = CAN A, red = CAN B, green = power. Each bus is one linear path with an isolation jumper (J_A / J_B) at every node (breaks both CAN_H and CAN_L). Termination is not shown: place the two 120 Ω terminators per bus on the bus side of the jumpers (R4.3).